

Hashing and Hash Tables

Where We Left Off

Finding one item among n :

Unsorted array, linear search $O(n)$

Sorted array, binary search $O(\log n)$

Today we get to **$O(1)$** average

Constant time - independent of n

The Problem

Our sensor readings have **names**, not numbers:

"oil_pressure"	1456 counts
"coolant_temp"	2011 counts
"fuel_level"	903 counts

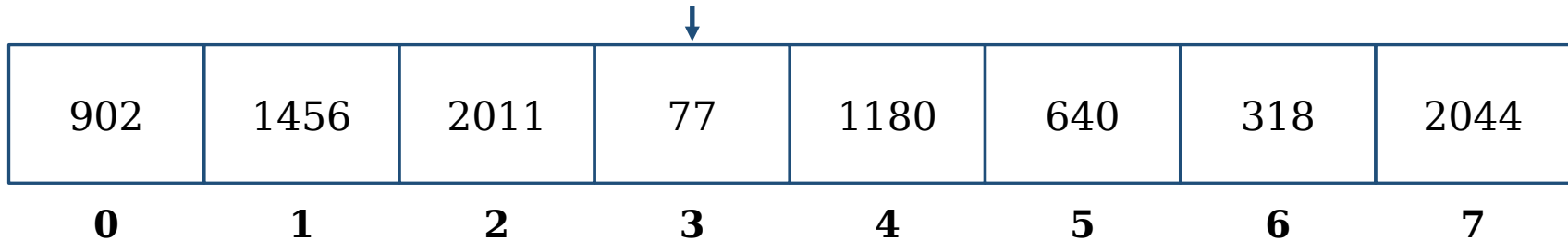
Given the name, find the reading

But an array index is a **number**

Direct Addressing

Sensor ID is already a small integer:

readings [3] is the reading for sensor 3



A diagram illustrating direct addressing. It shows a horizontal array of eight cells, each containing a numerical value. Below each cell is its corresponding index, from 0 to 7. A blue arrow points down to the cell at index 3, which contains the value 77. The values in the cells are: 902 at index 0, 1456 at index 1, 2011 at index 2, 77 at index 3, 1180 at index 4, 640 at index 5, 318 at index 6, and 2044 at index 7.

902	1456	2011	77	1180	640	318	2044
0	1	2	3	4	5	6	7

No hashing needed - the key IS the index, $O(1)$

Why Direct Addressing Fails

Keys are usually not small integers:

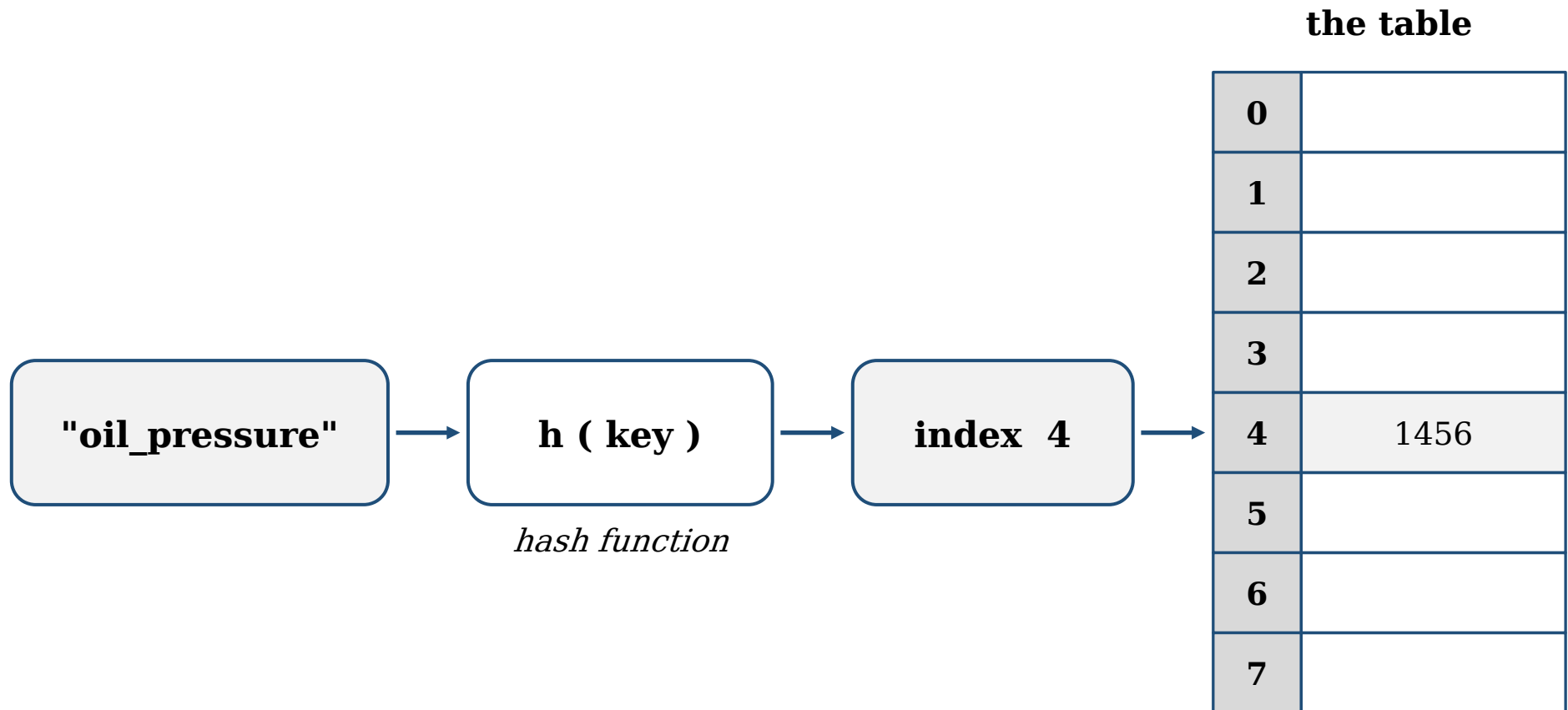
Strings "oil_pressure" is not an index

Sparse sensor IDs 1001, 5544, 90210
 three entries, 90,211 slots

Wide a 32-bit ID needs 4 billion slots

We need a way to **shrink** a key to an index

The Hashing Idea



We do not search for the slot - we **compute** it

What a Hash Function Is

A function that maps a key to an integer:

$$h : \text{key} \rightarrow 0 \dots m-1$$

m is the number of **buckets** in the table

Same key $\rightarrow$ **always** the same bucket

Different keys $\rightarrow$ **usually** different buckets

Properties of a Good Hash Function

Deterministic same input, same output

Fast cheaper than the search it replaces

Uniform spreads keys evenly over m

Avalanche one bit in changes many bits out

It is **not** encryption and **not** a checksum

The Division Method

The simplest useful hash:

$$\text{index} = k \% m$$

k is the key as an integer,
 m is the table size

One instruction, always in range

Everything depends on the choice of **m**

Why m Should Be Prime

If $m = 16$, then $k \% 16$ keeps only the low **4 bits** - the rest of the key is thrown away

$0x1234 \rightarrow 4$

$0xABC4 \rightarrow 4$

$0x0004 \rightarrow 4$

A **prime** m makes every bit of k matter

Hashing Strings: First Attempt

Add up the characters:

```
unsigned h = 0 ;  
while ( *s ) h += *s++ ;
```

"ab" $\rightarrow 97 + 98 = 195$

"ba" $\rightarrow 98 + 97 = 195$

Every **anagram** collides, and the range is tiny

djb2: A Better String Hash

```
unsigned long hash ( const char *s ) {  
    unsigned long h = 5381 ;  
    while ( *s )  
        h = h * 33 + ( unsigned char ) *s++ ;  
    return h ;  
}
```

Multiplying **carries position into the value**
now "ab" and "ba" differ

FNV-1a: Another Good Choice

```
unsigned hash ( const char *s ) {  
    unsigned h = 2166136261u ;  
    while ( *s ) {  
        h ^= ( unsigned char ) *s++ ;  
        h *= 16777619u ;  
    }  
    return h ;  
}
```

XOR then multiply - better avalanche than djb2

Hashing a Struct

Hash exactly the fields you **compare**

```
h = hash ( s.adc_counts ) ;  
h = h * 33 + hash ( s.timestamp_ms ) ;
```

Never hash the raw bytes of a struct:
padding bytes are uninitialized

Two equal structs would hash differently

The Golden Rule

If $a == b$

then $h(a) == h(b)$

Equality and hashing must **agree**

Break this and lookups silently fail

Collisions Are Unavoidable

Keys: billions of possible strings

Buckets: maybe 101

Pigeonhole principle

more pigeons than holes

→ some hole has two pigeons

A collision is **not** a bug

Handling it badly is

Worse Than You Expect

The **birthday paradox**

23 people in a room:

about a **50%** chance two share a birthday

Put $\sqrt{m}$ keys in m buckets:

a collision is already likely

365 buckets, 23 keys - plan for collisions

Two Ways to Handle Collisions

Separate chaining

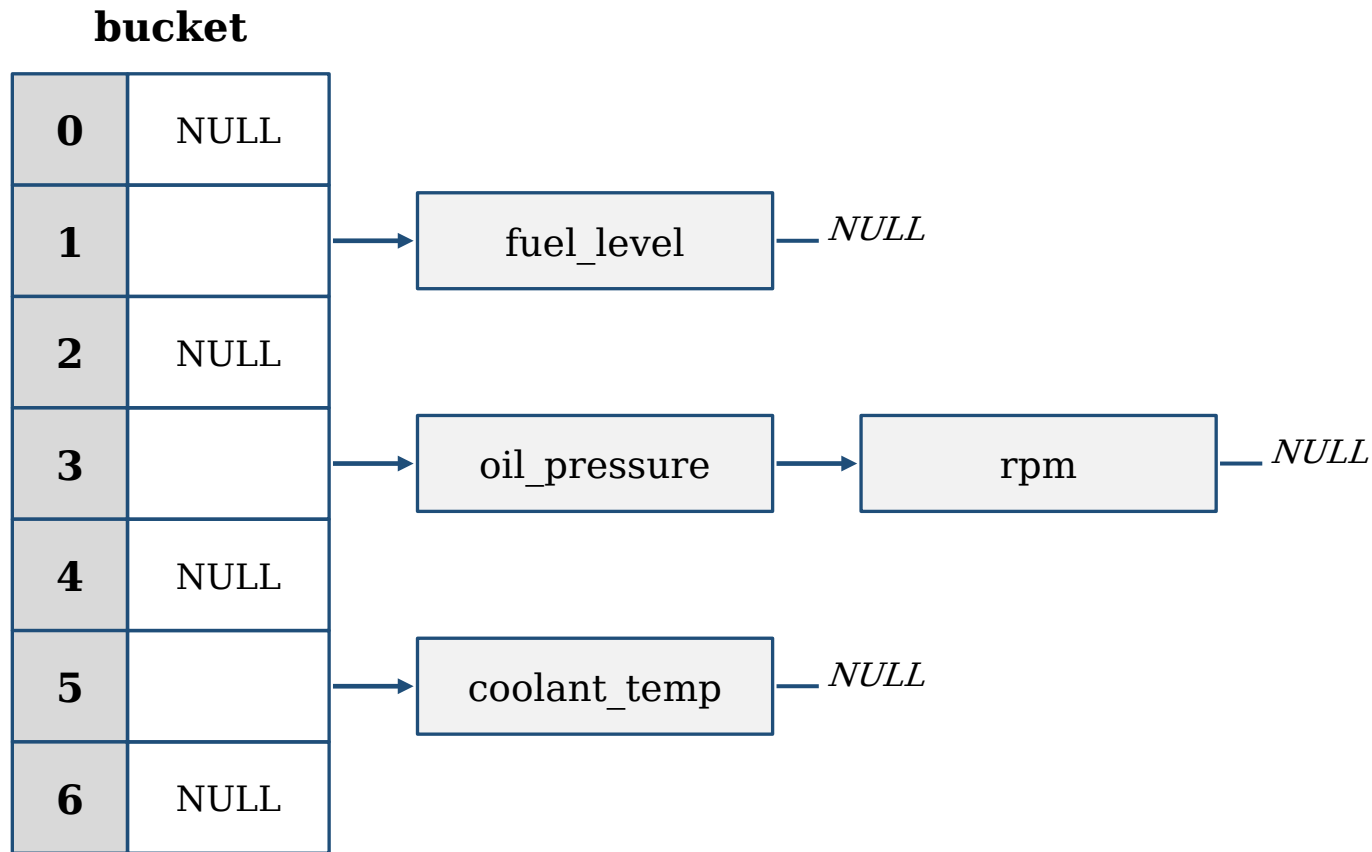
each bucket holds a **list** of entries

Open addressing

one entry per bucket; if full,
go look in another bucket

Every real hash table picks one of these

Separate Chaining



Each bucket holds a **list** of every key that hashed there

Chaining in C: The Entry

```
#define TABLE_SIZE      101

typedef struct entry {
    char      *key ;
    int       value ;
    struct entry *next ;
} entry_t ;

entry_t *tableG [ TABLE_SIZE ] = { NULL } ;
```

Chaining in C: Insert

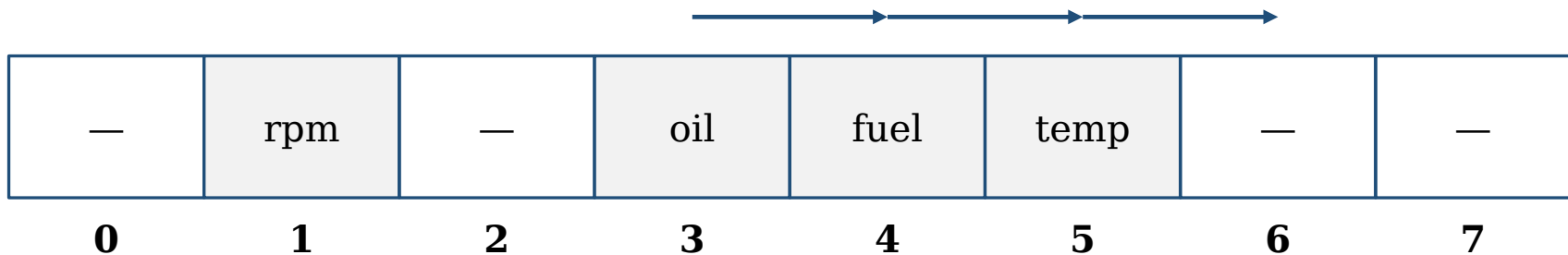
```
void insert ( const char *key, int value ) {  
    unsigned idx = hash ( key ) % TABLE_SIZE ;  
  
    entry_t *e = malloc ( sizeof ( entry_t ) ) ;  
    e->key      = strdup ( key ) ;  
    e->value    = value ;  
    e->next     = tableG [ idx ] ;  
    tableG [ idx ] = e ;  
}
```

Chaining in C: Lookup

```
int *lookup ( const char *key ) {  
    unsigned idx = hash ( key ) % TABLE_SIZE ;  
  
    for ( entry_t *e = tableG [ idx ] ; e ; e = e->next )  
        if ( strcmp ( e->key, key ) == 0 )  
            return &( e->value ) ;  
  
    return NULL ;                                /* not found */  
}
```

Open Addressing: Linear Probing

$h(\text{"speed"}) = 3$, but slot 3 is taken



Try 3, 4, 5, 6 ... take the **first empty slot**

Lookup repeats the same walk and stops at the first empty slot

Linear Probing in Code

```
unsigned idx = hash ( key ) % m ;

while ( table [ idx ].used ) {
    if ( strcmp ( table [ idx ].key, key ) == 0 )
        return idx ;                /* found */
    idx = ( idx + 1 ) % m ;          /* next slot */
}

return NOT_FOUND ;                 /* hit an empty slot */
```


Primary Clustering

Linear probing makes runs of full slots

A long run gets **longer**:

any key landing in it walks to the end

Two clusters that touch become one big one

Cost grows much faster than n does

Quadratic and Double Hashing

Linear $\text{idx} + 1, \text{idx} + 2, \text{idx} + 3 \dots$

Quadratic $\text{idx} + 1, \text{idx} + 4, \text{idx} + 9 \dots$
breaks up clusters

Double $\text{idx} + 1 \cdot h_2(k), \text{idx} + 2 \cdot h_2(k) \dots$
each key gets its own step size

h_2 must never return 0

Deletion and Tombstones

Open addressing cannot just empty a slot

Lookup stops at the first **empty** slot,
so an emptied slot **hides** everything behind it

Mark it **DELETED** instead - a **tombstone**
 probing walks past it
 insert may reuse it

Too many tombstones → rebuild the table

The Load Factor

$$\alpha = n / m$$

n = entries stored

m = buckets available

$\alpha = 0.5$ half full

$\alpha = 1.0$ as many entries as buckets

α is the one number that predicts speed

Cost of Chaining

Average list length = α

Successful lookup $\approx 1 + \alpha/2$ comparisons

Unsuccessful lookup $\approx 1 + \alpha$ comparisons

Both are $O(1 + \alpha)$

Keep α bounded and this is $O(1)$

Cost of Linear Probing

α	Chaining	Probing: hit	Probing: miss
0.25	1.25	1.2	1.4
0.50	1.50	1.5	2.5
0.75	1.75	2.5	8.5
0.90	1.90	5.5	50.5
0.95	1.95	10.5	200.5

Average slots touched per lookup

Linear probing falls apart as the table fills

The Practical Rule

Keep $\alpha \leq 0.7$

When the table passes that point:
allocate a **bigger** table (usually $2\times$)
rehash every entry into it
free the old table

The new m changes every index

Rehashing Is $O(n)$ - Once

One rehash costs $O(n)$

But it doubles the capacity, so it happens less and less often:

grow at 8, 16, 32, 64, 128 ...

Total cost of n inserts = $O(n)$

Each insert is $O(1)$ **amortized**

The Worst Case Is $O(n)$

If every key hashes to the **same** bucket:

chaining → one long linked list
probing → one long walk

Lookup becomes $O(n)$

A bad hash function turns a hash table into a linked list with extra steps

Hash Flooding

An attacker who knows h can craft keys that **all** land in one bucket

n requests $\rightarrow O(n^2)$ work $\rightarrow$ server stalls

The defence: a **random seed** per process
same key hashes differently each run

Rust does this by default; C++ does not

Complexity Summary

Operation	Average	Worst case
Insert	$O(1)$	$O(n)$
Lookup	$O(1)$	$O(n)$
Delete	$O(1)$	$O(n)$
Visit every entry	$O(n + m)$	$O(n + m)$
Space	$O(n)$	$O(n)$

Average assumes a good hash and α kept small

Compared With What We Know

Structure	Lookup	Sorted order	Extra memory
Unsorted array	$O(n)$	no	none
Sorted array	$O(\log n)$	yes	none
Balanced tree	$O(\log n)$	yes	pointers
Hash table	$O(1)$ avg	no	empty buckets

A hash table trades **ordering** for **speed**

Space Complexity

A hash table stores n entries in m buckets

$m > n$ always (we keep $\alpha \leq 0.7$)

Chaining	n entries + m pointers + n next pointers
Probing	m slots, about 30% of them empty

Still $O(n)$ - but with a real constant

Hash Tables in C

The C standard library has **no** hash table

Your choices:

write one (we just did)

uthash header-only macros

GLib GHashTable

POSIX hcreate / hsearch (one table only)

This is why C programmers know how these work

uthash in C

```
#include "uthash.h"
```

```
typedef struct reading {  
    char          name [ 32 ] ;      /* the key */  
    int           counts ;  
    UT_hash_handle hh ;              /* bookkeeping */  
} reading_t ;
```

```
reading_t *readingsG = NULL ;  
HASH_ADD_STR ( readingsG, name, r ) ;  
HASH_FIND_STR ( readingsG, "oil_pressure", r ) ;
```

C++: std::unordered_map

```
#include <unordered_map>  
#include <string>
```

```
std::unordered_map < std::string, int > readings ;
```

```
readings [ "oil_pressure" ] = 1456 ;  
readings [ "coolant_temp" ] = 2011 ;
```

```
int c = readings [ "oil_pressure" ] ;
```


C++: The [] Trap

```
if ( readings [ "rpm" ] == 0 ) ...
```

This **inserts** "rpm" with value 0

To look without inserting:

```
if ( readings.count ( "rpm" ) ) ...  
auto it = readings.find ( "rpm" );  
if ( it != readings.end ( ) ) use ( it->second );
```

C++20: `if ( readings.contains ( "rpm" ) ) ...`

C++: Iterating and Erasing

```
for ( const auto & [ name, counts ] : readings )  
    std::cout << name << " = " << counts << "\n" ;  
  
readings.erase ( "fuel_level" ) ;  
  
readings.reserve ( 1000 ) ;    /* avoid rehashing */  
  
readings.load_factor ( ) ;    /* current  $\alpha$  */  
readings.max_load_factor ( 0.75f ) ;
```

C++: `std::unordered_set`

```
#include <unordered_set>

std::unordered_set < int > seen ;

if ( ! seen.insert ( id ).second )
    std::cout << "duplicate id\n" ;
```

A map with keys and no values

This is the $O(n)$ duplicate finder from last lecture

C++: map vs unordered_map

	std::map	std::unordered_map
Structure	balanced tree	hash table
Lookup	$O(\log n)$	$O(1)$ average
Order	sorted by key	unspecified
Key needs	operator <	std::hash + ==
Rust name	BTreeMap	HashMap

Need keys in order? Use the tree.

C++: Hashing Your Own Type

```
struct key_t { int sensor ; int channel ; } ;

template < > struct std::hash < key_t > {
    size_t operator ( ) ( const key_t & k ) const {
        return std::hash<int>{} ( k.sensor ) * 33
            ^ std::hash<int>{} ( k.channel ) ;
    }
} ;
```

You must also supply **operator ==**

Rust: HashMap

```
use std::collections::HashMap ;

let mut readings : HashMap < String, u16 > =
    HashMap::new ( ) ;

readings.insert ( "oil_pressure".to_string ( ), 1456 ) ;

if let Some ( c ) = readings.get ( "oil_pressure" ) {
    println! ( "{} counts", c ) ;
}
```

Rust: The entry API

Count how many times each command arrives:

```
let mut counts : HashMap < & str, u32 > = HashMap::new  
( ) ;  
  
for cmd in log.iter ( ) {  
    *counts.entry ( cmd ).or_insert ( 0 ) += 1 ;  
}
```

Rust: Other entry Forms

```
map.entry ( k ).or_insert_with ( Vec::new ) ;
```

```
map.entry ( k ).or_default ( ) ;
```

```
map.entry ( k ).and_modify ( |v| *v += 1 )  
    .or_insert ( 1 ) ;
```

`or_insert_with` builds the value **only if needed**

Rust: HashSet

```
use std::collections::HashSet ;
```

```
let mut seen : HashSet < u32 > = HashSet::new ( ) ;
```

```
if ! seen.insert ( id ) {  
    println! ( "duplicate {}", id ) ;  
}
```

insert returns **true** if the value was new

Rust: Deriving Hash

```
#[ derive ( Hash, PartialEq, Eq ) ]  
struct Key {  
    sensor : u8,  
    channel : u8,  
}
```

Now `Key` can be a **key**

The compiler writes both, and they **agree**

Rust: Why f32 Cannot Be a Key

f32 and f64 do **not** implement Eq

NaN != NaN

A key that is not equal to itself can never be found again

Store the **raw counts**, not the psi
u16 is a key; f32 is not

Rust: Iteration Order

```
for ( name, counts ) in &readings {  
    println! ( "{}: {}", name, counts ) ;  
}
```

The order is **not defined**
and changes between runs

Need order? use **BTreeMap**
 or collect and sort

Rust: Choosing a Hasher

Default: **SipHash 1-3** with a random seed
safe against hash flooding
not the fastest

For trusted, internal keys:
use `rustc_hash::FxHashMap` ;
use `ahash::AHashMap` ;

Same API, different speed/safety trade

The Three Languages

	C	C++	Rust
Hash map	none	unordered_map	HashMap
Hash set	none	unordered_set	HashSet
Collisions	you choose	chaining	open addressing
Default hash	you write it	std::hash	SipHash 1-3
Random seed	no	no	yes
Ordered map	none	std::map	BTreeMap
Perfect hash	gperf	gperf / frozen	phf crate

C gives you nothing - which is why we built one

Perfect Hashing

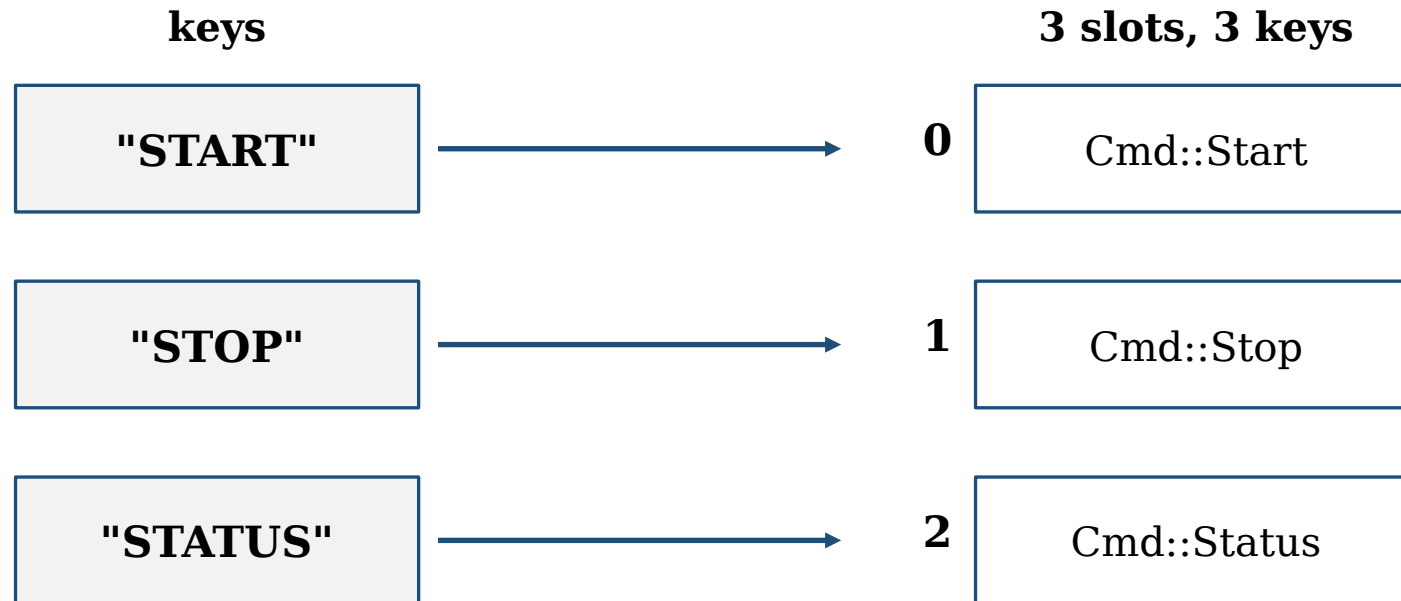
What if the set of keys is **known in advance**?

Reserved words, opcodes, command names,
HTTP headers, CSS properties

Then we can build a hash function with
no collisions at all

Lookup becomes **$O(1)$ worst case**

Perfect and Minimal



No collisions. Three keys, three slots - **minimal** perfect hashing
 $O(1)$ worst case, not just average

The Two Conditions

Perfect no two keys collide
 $m \geq n$

Minimal perfect perfect, and $m = n$
 every slot used, $\alpha = 1.0$

Both require the key set to be **fixed**

Add one key $\rightarrow$ build a new function

Where Perfect Hashing Is Used

Compilers is this word a keyword?

Assemblers instruction mnemonics

Web servers HTTP header names

Firmware command tables in flash

Small fixed sets, looked up constantly

gperf: Generating a Perfect Hash

Our three commands from the enums lecture:

```
%{  
#include "commands.h"  
%}  
%%  
START,          CMD_START  
STOP,           CMD_STOP  
STATUS,         CMD_STATUS  
%%
```

gperf: What It Generates

```
$ gperf -t commands.gperf > commands.c
```

```
static unsigned hash ( const char *s, size_t len )  
{  
    static const unsigned char asso [ ] = { ... } ;  
    return len + asso [ s [ 0 ] ] + asso [ s [ len-1 ] ] ;  
}
```

Two table lookups and two adds - **no loop**

gperf: Using the Result

```
const cmd_t *c = in_word_set ( buf, strlen ( buf ) ) ;  
  
if ( c != NULL )  
    dispatch ( c->code ) ;  
else  
    report ( ERR_UNKNOWN_COMMAND ) ;
```

You **still** compare the string once
an unknown word can hash to a valid slot

FKS: Perfect Hashing in Theory

Two levels:

Level 1 hash n keys into n buckets

Level 2 bucket with k keys gets its own
table of k^2 slots

k^2 slots $\rightarrow$ a collision-free h is easy to find

Total space is still **$O(n)$**

Rust: the phf crate

```
use phf::phf_map ;
```

```
static CMDS : phf::Map < & 'static str, Cmd > = phf_map! {  
    "START"    => Cmd::Start,  
    "STOP"     => Cmd::Stop,  
    "STATUS"   => Cmd::Status,  
};
```

Built at **compile time**, stored in **flash**

C++: Perfect Hashing

No standard tool, but the same options:

gperf	works on C++ too
frozen	constexpr maps, header-only
constexpr	build your own at compile time

```
constexpr frozen::unordered_map < ... > cmds = {  
    { "START", Cmd::Start }, ... } ;
```


Perfect vs Ordinary

	Ordinary hash	Perfect hash
Key set	unknown, grows	fixed, known
Collisions	expected	none
Lookup	$O(1)$ average	$O(1)$ worst case
Built	at run time	at build time
Add a key	just insert	regenerate
Memory	$\alpha \leq 0.7$, 30% spare	exactly n slots

Use a perfect hash when the keys ship **with the program**

Worked Example

A log of commands arrives from the ESP32:

STATUS START STATUS READ START STATUS

Count how many times each one appeared

How would you do it with an **array**?

Example in C++

```
std::unordered_map < std::string, int > counts ;
```

```
for ( const std::string & cmd : log )  
    counts [ cmd ] ++ ;
```

```
for ( const auto & [ cmd, n ] : counts )  
    std::cout << cmd << " " << n << "\n" ;
```

Here **[]** inserting **0** is exactly what we want

Example in Rust

```
let mut counts : HashMap < & str, u32 > = HashMap::new  
( ) ;
```

```
for cmd in log.iter ( ) {  
    *counts.entry ( cmd ).or_insert ( 0 ) += 1 ;  
}
```

```
for ( cmd, n ) in &counts {  
    println! ( "{} {}", cmd, n ) ;  
}
```

Example in C

```
for ( int i = 0 ; i < n_log ; i++ ) {  
    int *p = lookup ( log [ i ] ) ;  
    if ( p )          *p += 1 ;  
    else              insert ( log [ i ], 1 ) ;  
}
```

Our own table from earlier

Two lines in C++, seven in C, plus the table

Complexity of the Example

Array scan $O(n \cdot k)$
 $n \log$ entries, k known commands

Hash table **$O(n)$** average

Sort then count $O(n \log n)$

For $n = 100,000$ and $k = 20$:
 2 million compares vs 100,000

Seeing a Hash Table in the Debugger

Our C table is just an array of pointers:

```
(gdb) p tableG [ 42 ]  
(gdb) p *tableG [ 42 ]  
(gdb) p *tableG [ 42 ]->next
```

Walk the chain and you can **see** the collision

```
C++ and Rust:    p readings.size ( )  
                  p readings
```

Hash Tables on a Microcontroller

Ask before you reach for one:

How many keys? under 10 → a linear scan wins

Fixed keys? → perfect hash in flash

Is there a heap? chaining calls malloc

An ESP32-S3 has RAM measured in **kilobytes**

Choosing a Structure

If you need ...	Use
Key is a small integer	a plain array
Fewer than about 10 keys	a linear scan
Fixed keys, known at build	a perfect hash
Fast lookup by any key	a hash table
Keys in sorted order	a tree: map or BTreeMap
Ranges or nearest match	a tree
Membership only	a hash set

Common Mistakes

Changing a key **after** it is inserted
the entry is now in the wrong bucket

Hashing and comparing **different** fields

Using `[]` in C++ to test for a key

Letting α reach 1.0 with open addressing

Assuming the iteration order means anything

What We Can Now Do

Compute a bucket instead of searching

Choose a hash function and a table size

Handle collisions by chaining or probing

Predict cost from the load factor α

Generate a perfect hash for fixed keys

Let's practice

Exercise: Using **hashpractice**